INORGANIC CUMULATIVE EXAM

February 8, 1997 .

1) The molecule PF_5 is a trigonal bipyramid in all phases; however, separate ^{19}F resonances for the axial and equatorial fluorines are not observed at room temperature? *Explain.*

2) Phosphorus pentachloride is molecular in the gas phase and pure liquid; however, MeCN solution are conducting. Two different anions (one being Cl^-) are proposed to explain these observations. *Account for the electrical conductivity of the MeCN solutions.*

3) Both PF_3 and PCl_3 can displace CO from $Ni(CO)_4$. *Describe the nature of the Ni-PX_3 bonding to account for these displacement reactions.*

4) The methanolysis of PCl_3 gives $P(OMe)_3$ but the hydrolysis of PCl_3 can give two different isomeric products. *Suggest structures for these two products of hydrolysis.*

5) Antimony pentafluoride is associated in the gas phase, forms a viscous (apparently polymeric) liquid, and is tetrameric in the solid. *What type of bonding interaction could account for all these observations?.*

6) Discuss the probable products for the very rapid reaction between PCl_3 and PBr_3. *Suggest a mechanism to account for the reaction.*

$$PCl_3 + PBr_3 \longrightarrow ???$$

7) The compound with the *empirical formula* PF_2 is stable and well characterized. *Explain*

8) Antimony forms several halogenated anions including SbF_4^- and $SbCl_6^-$. *Suggest structures for each of these anions.*